

Sensor Placement for Output-Only Identification: A Mini Literature Review

Summary for supervision discussion

This note summarises a quick literature review undertaken to answer one specific question before building Level 2C: *given a maximum of four sensors, what criterion should be used to choose where to place them, and is that criterion actually appropriate for an output-only, SSI-based identification problem rather than a generic state-estimation problem?*. Background concepts are introduced from scratch where they may not already be familiar. Findings are marked in blue (background), green (verified results this review relies on), and red (genuinely open questions Level 2C should investigate).

1 Background: what observability-based placement actually means

1.1 The state-space picture, and why placement is a genuine question

Recall the discrete-time state-space model underlying SSI itself:

$$x_{k+1} = Ax_k, \quad y_k = Cx_k,$$

(noise omitted here for clarity). x_k is the internal state (in the SSI context, ultimately related to the floe's modal amplitudes); y_k is what is actually measured. Different sensor locations correspond to different *rows* of C : each sensor contributes one row, since C maps the full internal state down to whatever a specific physical location on the floe actually feels. Choosing where to place four sensors is therefore, mathematically, choosing which four rows of a much larger candidate C matrix to keep. This is the single unifying idea behind every paper reviewed below.

Observability. A system is observable if its entire internal state x_k can, in principle, be reconstructed exactly from a long enough record of y_k alone. A system can be observable in a binary yes/no sense, but real, noisy, finite-length data makes a stronger question more useful: *how well* is each internal direction of x_k observed, not just whether it is observable at all in the idealised infinite-data limit.

1.2 Two genuinely different kinds of observability Gramian

Finite-horizon Gramian. Over a fixed number of time steps N (or a fixed continuous duration T), the observability Gramian is

$$W_o^{(N)} = \sum_{k=0}^{N-1} (A^k)^T C^T C A^k \quad (\text{discrete}), \quad W_o(T) = \int_0^T e^{A^T t} C^T C e^{At} dt \quad (\text{continuous}).$$

Both are finite sums or finite integrals: they are well-defined for *any* A , regardless of where its eigenvalues sit, since there is no limiting process involved.

Infinite-horizon Gramian. Letting $N \rightarrow \infty$ or $T \rightarrow \infty$ gives the *general* (as opposed to interval-specific) observability Gramian, conventionally computed not by literally summing infinitely many terms but by solving a Lyapunov equation directly:

$$A^T W_o A - W_o + C^T C = 0 \quad (\text{discrete}), \quad A^T W_o + W_o A + C^T C = 0 \quad (\text{continuous}).$$

This is convenient exactly because it avoids the infinite sum or integral, but the convenience comes at a real cost: **the Lyapunov equation only has a valid solution if A is strictly stable** (discrete: every eigenvalue strictly inside the unit circle, $|\lambda| < 1$; continuous: every eigenvalue with strictly negative real part). If any eigenvalue sits on, or outside, that boundary, the equation has no solution (or an ill-conditioned one), and the Gramian is not meaningfully computable this way.

Why this distinction is directly important for this project. Level 1 of this project's own validation ladder found that a purely forced tone produces $\zeta \approx 0$ to within numerical precision:

eigenvalues sitting essentially exactly on the unit circle (discrete) or imaginary axis (continuous), not strictly inside either stability region. Any placement criterion built on an *infinite-horizon*, Lyapunov-equation-based Gramian is therefore not straightforwardly applicable to lightly-damped floe dynamics of exactly this kind. As Section 2 below shows, this is not a minor edge case in the literature reviewed here - it affects three of the five papers directly.

1.3 Scalar measures of the Gramian, and classical optimal-design terminology

W_o (of either kind above) is a matrix, not a single number, so comparing candidate sensor layouts requires reducing it to a scalar. Four measures recur throughout the literature reviewed:

Measure	Classical name	What it rewards
$\det(W_o)$	D-optimality	Overall "volume" of observability across all directions
$\text{tr}(W_o)$	A-optimality (related)	Total observed energy, summed over directions
$\lambda_{\min}(W_o)$	E-optimality	The single worst-observed direction specifically
$\kappa(W_o) = \lambda_{\max}/\lambda_{\min}$	Condition number	How evenly-balanced observability is across directions

Why these are genuinely different questions, not just different formulas. The determinant is the *product* of all eigenvalues of W_o : if even one direction is very poorly observed, the determinant collapses toward zero regardless of how well every other direction is observed, so it can be a poor discriminator when observability is already weak overall. The trace is a *sum*: a configuration can have a large trace purely by observing one direction extremely well, while leaving another direction almost completely unobserved, because a sum does not penalise a single near-zero term the way a product does. The minimum eigenvalue asks a narrower, more targeted question: forget the average, what is the worst case? Condition number is a relative measure of balance, independent of overall scale.

1.4 Modular and submodular set functions

Modular and submodular set functions. Choosing a subset of sensors from a larger candidate set is a *combinatorial* optimisation problem: for n candidate locations and a desired 4 sensors, there are $\binom{n}{4}$ possible configurations, which grows extremely quickly and is generally intractable to search exhaustively once n is more than a handful. A set function F (here, a Gramian-based placement score) is *modular* if adding one more sensor always contributes exactly the same amount to F , regardless of which other sensors are already selected - in this special case the optimal subset can be found essentially by sorting candidates individually and taking the top few, with no combinatorial search needed at all. F is *submodular* if adding a sensor helps less and less as more sensors are already present (diminishing returns) - a weaker, more common property that still permits a simple, efficient *greedy* algorithm (repeatedly add whichever single remaining sensor improves F the most) to get provably close to the true optimum.

2 Papers reviewed, verified against primary sources

2.1 Geelen, Yntema, Molenaar & Keesman (2021)

Optimal Sensor Placement in Hydraulic Conduit Networks: A State-Space Approach, Water, 13(21), 3105.

Verified directly. Eq. (10) of the paper is exactly the continuous, infinite-horizon Gramian $W_o = \int_0^\infty e^{A^T \tau} C^T C e^{A \tau} d\tau$, computed by solving the continuous Lyapunov equation $A^T W_o + W_o A = -C^T C$. The paper is explicit, not merely implicit, about the requirement this carries: it states directly that the network "system is called (asymptotically) stable," and only for such a system can W_o be calculated this way. The paper chooses $\lambda_{\min}(W_o)$ (E-optimality) as its placement criterion, explicitly interpreting W_o as a Fisher Information Matrix and citing the eigenvalue-optimality criterion of Georges (1995).

Relevance. The $\lambda_{\min}$ criterion itself, independent of the infinite-horizon formulation it happens to be paired with here, is directly useful: it asks exactly the question this project cares about - not "is the overall configuration informative" but "is there a specific dynamic direction (e.g. the difference between two close candidate frequencies) that remains poorly observed no matter how strong the total response is." The Gramian form itself, however, should not be transplanted directly (see Section 1.2).

2.2 Summers & Lygeros (2013/2014)

Optimal Sensor and Actuator Placement in Complex Dynamical Networks, arXiv:1306.2491 (also IFAC Proceedings 2014).

Verified directly. The paper's controllability Gramian (observability is treated as the dual property throughout) is explicitly defined at a finite time t : $W_c(t) = \int_0^t e^{A\tau} B B^T e^{A^T \tau} d\tau$, with no infinite-horizon limit taken anywhere in the placement-theoretic results. This Gramian is well-defined for any finite t and any A , with no stability requirement - it is, of the papers reviewed, the cleanest example of the finite-horizon form. The paper's central theoretical result is that any linear function of this Gramian (in particular, trace) is a *modular* set function over the choice of actuator/sensor placements, meaning the placement problem for such metrics can be solved essentially by ranking candidates individually, with no combinatorial search needed.

Relevance. Establishes that placement is a genuine, well-posed optimisation problem, and that trace-type objectives are the cheapest to optimise. Its finite-time Gramian is also a concrete template for exactly the modification Level 2C needs to make to Geelen's, Yamada's, and Bartos's placement criteria (Section 3).

2.3 Yamada, Sasaki, Nagata, Nakai, Tsubakino & Nonomura (2023)

Efficient Sensor Node Selection for Observability Gramian Optimization, Sensors, 23(13), 5961.

NOTE The defining equation (Eq. 3) is

$$W_o = \sum_{k=0}^{\infty} (A^k)^T C^T C A^k,$$

an explicitly *infinite* sum, evaluated in practice by solving the discrete-time Lyapunov equation $A^T W_o(I_p)A - W_o(I_p) + (PC)^T(PC) = 0$ (Eq. 9). This carries exactly the same requirement as Geelen's continuous case, translated to discrete time: every eigenvalue of A must lie strictly inside the unit circle. A discrete system with an eigenvalue on the unit circle (again, exactly the Level 1 forced-tone case) is not compatible with this formulation either.

What remains genuinely useful. The $\log \det(W_o)$ criterion (D-optimality, Eq. 10), and the four solution algorithms compared (SDP, an approximate convex/Newton relaxation, pure greedy, gradient-greedy), are independent of which Gramian (finite- or infinite-horizon) is plugged in. The paper reports that approximate greedy selection is particularly effective when the number of sensors to select is comparable in size to the reduced system's own state dimension, a regime plausibly matching this project's four-sensor, low-order case.

2.4 Bartos & Kerkez (2021)

Observability-Based Sensor Placement Improves Contaminant Tracing in River Networks, Water Resources Research, 57(7).

NOTE The paper explicitly defines a finite-interval Gramian $W_o(t_f, t_0)$ (Eq. 15) for a general interval, but its *actual placement algorithm*, used for the reported trace- and rank-based results, works with the "infinite time-horizon Observability Gramian," stated explicitly to be obtained "by letting t_f approach infinity," and computed via the discrete-time Lyapunov equation $A^T W_o A -$

$W_o + C^T C = 0$ (Eq. 16). This is the same Lyapunov-equation form as Yamada’s, carrying the same strict-discrete-stability requirement. The paper’s own motivation for choosing the infinite-horizon form is informative: it is explicitly intended for systems whose dynamics ”may take an indefinitely long time to reach a steady state” - i.e. built specifically for genuinely decaying systems observed over a long window, not for lightly-damped or marginal dynamics.

Numerical Claims (i) The paper does evaluate reconstruction error over ”200 randomized initial contaminant loads” . (ii) The paper states directly: ”the mean-squared error associated with the trace-optimized strategy is generally larger than the error associated with randomized sensor placements” (confirmed verbatim) - trace-based placement genuinely underperformed a random baseline in this study. A further, more specific result: rank-based placement required only 32 of 223 candidate river segments (about 14%) to achieve complete observability, versus 213 of 223 for trace-based placement.

Relevance - the strongest direct link to this project’s own results. The paper’s own framing of this contrast, verified directly: ”trace-optimized sensor placement does not encourage sufficient exploration of the state space... placing sensors evenly throughout the network (i.e., exploration) does more to reduce overall reconstruction error than focusing all sensors on the most information-dense regions (i.e., exploitation).” This maps closely onto the Level 2B finding: the radial-spread sensor configuration performed far worse (cross-MAC 0.86) than the edge-clustered configurations (0.13-0.27) at separating two known frequencies. A plausible interpretation, motivated directly by this paper, is that radial spread behaved like a trace-favoured (exploitative) layout, sampling large-signal but spatially redundant information, while the edge-based layouts behaved more like a rank-favoured (exploratory) layout. This is an interpretation of the existing Level 2B result, not something Bartos’s paper itself demonstrates for a hydroelastic system, and (per the correction above) not directly transferable at the level of the Gramian formulation itself.

2.5 Qi, Sun & Kang (2014-2016)

Adaptive Optimal PMU Placement Based on Empirical Observability Gramian, arXiv:1411.7016 (journal version: IEEE Transactions on Power Systems, 30(4), 2015, as *Optimal PMU Placement for Power System Dynamic State Estimation by Using Empirical Observability Gramian*).

Verified directly, and structurally the most compatible Gramian form of the three Gramian-based papers. The paper’s continuous theoretical definition (Eq. 10) is written with an infinite-time integral, but the discrete form actually used for computation (Eq. 11) is

$$W_o = \sum_{l=1}^r \sum_{m=1}^s \frac{1}{r s c_m^2} \sum_{k=0}^K (T^l \Psi_k^{lm} T^l) \Delta t_k,$$

an explicit, genuinely *finite* sum over $k = 0, \dots, K$, where K is set by the actual simulation duration used. Unlike Geelen’s, Yamada’s, and Bartos’s placement Gramians, this one carries no stability requirement at all: it is a direct numerical construction from perturbed, finite-duration simulated trajectories, not a Lyapunov-equation shortcut. This also means it does not require an explicit closed-form linear (A, C) model - useful if the underlying system is only available as a nonlinear simulator.

Main finding: no universally best scalar measure. Verified directly: when few sensors are available and observability is weak overall (directly analogous to this project’s four-sensor constraint), $\lambda_{\min}$ and condition number outperform the determinant, because the determinant’s product structure makes it collapse toward zero (and hence stop discriminating usefully between candidate layouts) once even one direction is weakly observed. Once enough sensors are available for observability to be reasonably strong overall, the determinant becomes a good summary measure. The paper’s own proposed solution is an *adaptive* criterion: use $\lambda_{\min}$ or condition number when

observability is weak, transition to determinant once it is not. The correct authors, verified directly, are Qi, Sun, and Kang, correcting an earlier misattribution to "Taha et al."

Relevance. Given this project's hard four-sensor ceiling, the weak-observability regime this paper specifically studies is exactly the relevant one. Combined with its genuinely finite (stability-agnostic) Gramian construction, this paper is now, after direct verification, the single most structurally appropriate template of the five for this project's own Gramian computation.

3 Synthesis

Paper	Gramian form (verified)	Criterion	Stability requirement
Geelen et al. (2021)	Continuous, infinite-horizon (Lyapunov)	$\lambda_{\min}$	Strictly stable ($\text{Re}(\lambda) < 0$), stated explicitly in the paper
Summers & Lygeros	Continuous, explicitly finite-time $W_c(t)$	Linear trace (e.g.)	None - valid for any A
Yamada et al. (2023)	Discrete, infinite-horizon (Lyapunov)	log det	Strictly stable ($ \lambda < 1$) - not finite-horizon, contrary to an earlier draft of this review
Bartos & Kerkez (2021)	Discrete, infinite-horizon (Lyapunov)	Trace vs. rank	Strictly stable ($ \lambda < 1$) - not finite-horizon, contrary to an earlier draft of this review
Qi, Sun & Kang	Discrete, genuinely finite sum ($k = 0, \dots, K$)	All four, compared	None - valid for any A

The corrected central conclusion of this review. Of the five papers, only Summers & Lygeros (by explicit construction) and Qi, Sun & Kang (in their actual implemented discrete form) use a Gramian that is valid regardless of the stability of the recovered dynamics. Geelen, Yamada, and Bartos all rely on a Lyapunov-equation shortcut that requires strict stability, and none of the three is directly applicable, as written, to the lightly-damped or marginally-stable dynamics this project has already demonstrated (Level 1, $\zeta \approx 0$). This is a stronger and more specific conclusion than an earlier draft of this review reached, which incorrectly credited Yamada and Bartos with already using a stability-agnostic form.

The gap none of these papers close, and exactly what Level 2C should investigate. Every paper reviewed validates its chosen Gramian criterion against *state reconstruction accuracy* or dynamic state estimation error. None of them validate a Gramian criterion against *output-only, subspace-identification-based modal separation*, the actual quantity this project cares about and already measures directly in Level 2B via cross-MAC. Whether a Gramian-optimal sensor configuration is also an SSI-modal-separation-optimal configuration remains an open, unanswered question in this literature.

4 Proposed design for Level 2C

4.1 Building a state-space model that genuinely matches the generated signal

Two known frequencies ω_1, ω_2 (the 2.70/6.00 rad/s pair already established as spatially distinguishable in Levels 2A/2B is the natural continuity choice) define a real, exactly-known 4×4 state matrix, built directly from the confirmed Level 1 result that a forced tone has $\zeta \approx 0$ (eigenvalues on the unit circle):

$$A = \text{blkdiag}(R(\omega_1 \Delta t), R(\omega_2 \Delta t)), \quad R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

No SSI is involved in constructing A : this avoids the circularity of needing sensor placement to run SSI, while needing SSI's output to choose sensor placement.

NOTE Each candidate sensor’s row of C must have four real entries, matching the 4-dimensional real state above, not the raw 2-column complex spatial response $[\phi_1(\mathbf{r}_j), \phi_2(\mathbf{r}_j)]$. The correct mapping follows directly from $\text{Re}\{\phi e^{i\omega t}\} = \text{Re}(\phi) \cos(\omega t) - \text{Im}(\phi) \sin(\omega t)$: with state $x = [\cos \omega_1 t, \sin \omega_1 t, \cos \omega_2 t, \sin \omega_2 t]$, each candidate row must be

$$C_j = [\text{Re}(\phi_1(\mathbf{r}_j)), -\text{Im}(\phi_1(\mathbf{r}_j)), \text{Re}(\phi_2(\mathbf{r}_j)), -\text{Im}(\phi_2(\mathbf{r}_j))],$$

using the already-validated `evaluateDeflection` to obtain ϕ_1, ϕ_2 at each candidate location. This makes the Gramian’s underlying state-space model mathematically identical to the actual signal `syntheticSensorData` would generate for that configuration, not merely analogous to it.

4.2 Modal field normalisation: an explicit decision, not a silent default

Decision: normalise the two modal fields before constructing C , for this experiment. If ϕ_1 is naturally much larger in amplitude than ϕ_2 across the candidate grid, an unnormalised Gramian would partly reward “measures the already-large component well” rather than the geometric distinguishability Level 2C is meant to isolate – silently reintroducing the amplitude-imbalance question Level 3B already investigates separately. This experiment therefore uses

$$\tilde{\phi}_i(\mathbf{r}) = \frac{\phi_i(\mathbf{r})}{\|\phi_i\|_{\text{candidate grid}}}, \quad i = 1, 2,$$

before forming any C_j . An unnormalised, physics-weighted variant (retaining true relative amplitude) is a legitimate follow-up experiment, not run here, since it would re-couple exactly the two effects (geometry, amplitude) this design deliberately separates.

4.3 Explicit finite horizon, and what “convergence” means for an undamped system

The Gramian is computed as the explicit finite sum from Section 1.2,

$$W_o^{(N)}(\mathcal{S}) = \sum_{k=0}^{N-1} (A^k)^T C(\mathcal{S})^T C(\mathcal{S}) A^k,$$

for a chosen horizon N .

N requires its own sensitivity check, and “convergence” means something specific here. Because A ’s eigenvalues sit exactly on the unit circle by construction (an undamped system), $W_o^{(N)}$ cannot converge to a fixed matrix as $N \rightarrow \infty$: each cycle keeps adding observable energy, with nothing to bound the sum. The correct check is therefore not “does $W_o^{(N)}$ converge” (it structurally cannot), but whether the *ranking* of candidate configurations under each metric stabilises once N spans a sufficient number of oscillation cycles. This should be verified directly (e.g. compare rankings at N corresponding to 1, 2, 5, and 10 cycles of the slower frequency) before trusting a single chosen N .

4.4 Which selection method is licensed for which objective

Objective	Structural guarantee	Selection method
$\text{tr}(W_o)$	Modular (Summers & Lygeros)	Exact: rank candidates individually
$\log \det(W_o)$	Submodular under stated conditions (Yamada et al.)	Greedy or SDP relaxation
$\lambda_{\min}(W_o)$	No general guarantee	Greedy, used explicitly as a heuristic only
$\kappa(W_o)$	No general guarantee	Greedy, used explicitly as a heuristic only

Numerically, $\log \det(W_o)$ (via a stable `slogdet`-type routine) is computed directly rather than $\det(W_o)$ itself, and $\kappa(W_o)$ is *minimised* (equivalently, $1/\kappa$ or $-\kappa$ is maximised), since a well-conditioned, evenly-observed configuration corresponds to a small condition number, the opposite direction from the other three objectives.

4.5 Full procedure

1. Fix $(\omega_1, \omega_2) = (2.70, 6.00)$ rad/s and Δt ; construct the exact real 4×4 oscillator A .

2. Evaluate ϕ_1, ϕ_2 over a candidate spatial grid via `evaluateDeflection`.
3. Normalise both modal fields across the candidate grid (Section 1.3).
4. Construct each candidate row $C_j = [\text{Re}(\tilde{\phi}_1), -\text{Im}(\tilde{\phi}_1), \text{Re}(\tilde{\phi}_2), -\text{Im}(\tilde{\phi}_2)]$.
5. Verify ranking stability across N (Section 1.4) before fixing a working horizon.
6. For every candidate four-sensor set considered, compute $W_o^{(N)}$ and all four scalar objectives (Section 1.5), using exact or greedy selection as licensed per objective.
7. Run the full Forward-Model $\rightarrow$ `syntheticSensorData` $\rightarrow$ SSI pipeline only on the shortlisted finalists (one per objective), not the whole candidate grid.
8. For each finalist, distinguish between true spatial distinguishability and SSI recovery. First compute the Forward-Model cross-MAC, $\text{MAC}_{\text{FM},\text{cross}} = \text{MAC}(\Phi_{1,\text{FM}}, \Phi_{2,\text{FM}})$, which measures the intrinsic spatial distinguishability of the two true responses at the selected sensor locations. Then compute the SSI-to-truth MAC for each identified mode, $\text{MAC}_{\text{SSI} \rightarrow \text{FM},i} = \text{MAC}(\hat{\Phi}_i, \Phi_{i,\text{FM}})$, $i = 1, 2$, which measures how accurately SSI recovered each true spatial response. Report $\text{MAC}_{\text{FM},\text{cross}}$, $\text{MAC}_{\text{SSI} \rightarrow \text{FM},1}$, $\text{MAC}_{\text{SSI} \rightarrow \text{FM},2}$, $|\hat{\omega}_i - \omega_i|$ for each mode, and branch persistence *separately* – do not collapse these into one combined score. This gives the interpretation chain Gramian score $\rightarrow$ true spatial distinguishability $\rightarrow$ SSI recovery.
9. Compare which objective’s finalist actually performed best under real SSI-based identification, and report agreement or disagreement with the theoretical Gramian rankings honestly – a genuine disagreement is itself a valuable, reportable finding.